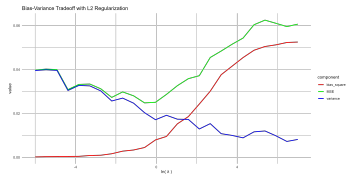
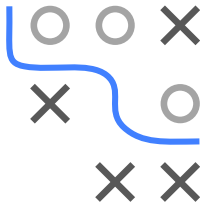


Introduction to Machine Learning

Regularization

Bias-Variance Decomposition for Ridge Regression (Deep-Dive)



Learning goals

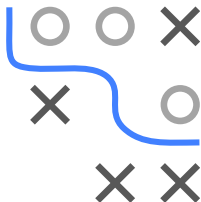
- Bias-Variance trade-off for ridge regression

BIAS-VARIANCE DECOMPOSITION FOR RIDGE I

For a linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$ with fixed design

$\mathbf{X} \in \mathbb{R}^{n \times p}$ and $\boldsymbol{\varepsilon} \sim (\mathbf{0}, \sigma^2 \mathbf{I}_n)$, bias of ridge estimator $\hat{\boldsymbol{\theta}}_{\text{ridge}}$ is given by

$$\begin{aligned}\text{Bias}(\hat{\boldsymbol{\theta}}_{\text{ridge}}) &:= \mathbb{E}[\hat{\boldsymbol{\theta}}_{\text{ridge}} - \boldsymbol{\theta}] = \mathbb{E}[(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{y}] - \boldsymbol{\theta} \\ &= \mathbb{E}[(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top (\mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon})] - \boldsymbol{\theta} \\ &= (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{X}\boldsymbol{\theta} + (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \underbrace{\mathbb{E}[\boldsymbol{\varepsilon}]}_{=0} - \boldsymbol{\theta} \\ &= (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{X}\boldsymbol{\theta} - \boldsymbol{\theta} \\ &= \left[(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} - (\mathbf{X}^\top \mathbf{X})^{-1} \right] \mathbf{X}^\top \mathbf{X}\boldsymbol{\theta}\end{aligned}$$



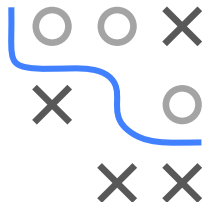
- Last expression shows bias of ridge estimator only vanishes for $\lambda = 0$, which is simply (unbiased) OLS solution
- It follows $\|\text{Bias}(\hat{\boldsymbol{\theta}}_{\text{ridge}})\|_2^2 > 0$ for all $\lambda > 0$

BIAS-VARIANCE DECOMPOSITION FOR RIDGE II

For the variance of $\hat{\theta}_{\text{ridge}}$, we have

$$\begin{aligned}\text{Var}(\hat{\theta}_{\text{ridge}}) &= \text{Var}\left((\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{y}\right) \quad \bigg| \quad \text{apply } \text{Var}_u(\mathbf{A}u) = \mathbf{A} \text{Var}(u) \mathbf{A}^\top \\ &= (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \text{Var}(\mathbf{y}) \left((\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top\right)^\top \\ &= (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \text{Var}(\epsilon) \mathbf{X} (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \\ &= (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \sigma^2 \mathbf{I}_n \mathbf{X} (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1}\end{aligned}$$

- $\text{Var}(\hat{\theta}_{\text{ridge}})$ is strictly smaller than $\text{Var}(\hat{\theta}_{\text{OLS}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ for any $\lambda > 0$, meaning matrix of their difference $\text{Var}(\hat{\theta}_{\text{OLS}}) - \text{Var}(\hat{\theta}_{\text{ridge}})$ is positive definite (bit tedious derivation)
- This further means $\text{trace}(\text{Var}(\hat{\theta}_{\text{OLS}}) - \text{Var}(\hat{\theta}_{\text{ridge}})) > 0 \forall \lambda > 0$



BIAS-VARIANCE DECOMPOSITION FOR RIDGE III

Having obtained the bias and variance of the ridge estimator, we can decompose its mean squared error as follows:

$$\text{MSE}(\hat{\theta}_{\text{ridge}}) = \|\text{Bias}(\hat{\theta}_{\text{ridge}})\|_2^2 + \text{trace}(\text{Var}(\hat{\theta}_{\text{ridge}}))$$

Comparing MSEs of $\hat{\theta}_{\text{ridge}}$ and $\hat{\theta}_{\text{OLS}}$ and using $\text{Bias}(\hat{\theta}_{\text{OLS}}) = 0$ we find

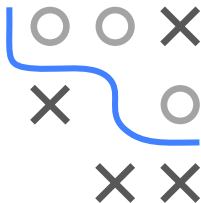
$$\text{MSE}(\hat{\theta}_{\text{OLS}}) - \text{MSE}(\hat{\theta}_{\text{ridge}}) = \underbrace{\text{trace}(\text{Var}(\hat{\theta}_{\text{OLS}}) - \text{Var}(\hat{\theta}_{\text{ridge}}))}_{>0} - \underbrace{\|\text{Bias}(\hat{\theta}_{\text{ridge}})\|_2^2}_{>0}$$

Since both terms are positive, sign of their diff is *a priori* undetermined.

► Theobald 1974 and ► Farebrother 1976 prove there always exists some $\lambda^* > 0$ so that

$$\text{MSE}(\hat{\theta}_{\text{OLS}}) - \text{MSE}(\hat{\theta}_{\text{ridge}}) > 0$$

Important theoretical result: While Gauss-Markov guarantees $\hat{\theta}_{\text{OLS}}$ is best linear unbiased estimator (BLUE), there are biased estimators with lower MSE.



BIAS-VARIANCE IN PREDICTIONS FOR RIDGE I

In supervised learning, our goal is typically not to learn an unknown parameter θ , but to learn a function $f(\mathbf{x})$ that can predict y given $\mathbf{x}$.

The bias and variance of predictions $\hat{f} := \hat{f}(\mathbf{x}) = \hat{\theta}_{\text{ridge}}^\top \mathbf{x}$ is obtained as:

$$\begin{aligned}\text{Bias}(\hat{f}) &= \mathbb{E}[\hat{f} - f] = \mathbb{E}[\hat{\theta}_{\text{ridge}}^\top \mathbf{x} - \theta^\top \mathbf{x}] = \mathbb{E}[\hat{\theta}_{\text{ridge}} - \theta]^\top \mathbf{x} \\ &= \text{Bias}(\hat{\theta}_{\text{ridge}})^\top \mathbf{x} \\ \text{Var}(\hat{f}) &= \text{Var}(\hat{\theta}_{\text{ridge}}^\top \mathbf{x}) = \mathbf{x}^\top \text{Var}(\hat{\theta}_{\text{ridge}}) \mathbf{x}\end{aligned}$$

The MSE of $\hat{f}$ given a fresh sample $(y, \mathbf{x})$ can now be decomposed as

$$\text{MSE}(\hat{f}) = \mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \text{Bias}^2(\hat{f}) + \text{Var}(\hat{f}) + \sigma^2$$

This decomposition is similar to the statistical inference setting before, however, the irreducible error σ^2 only appears for predictions as an artifact of the noise in the test sample.

